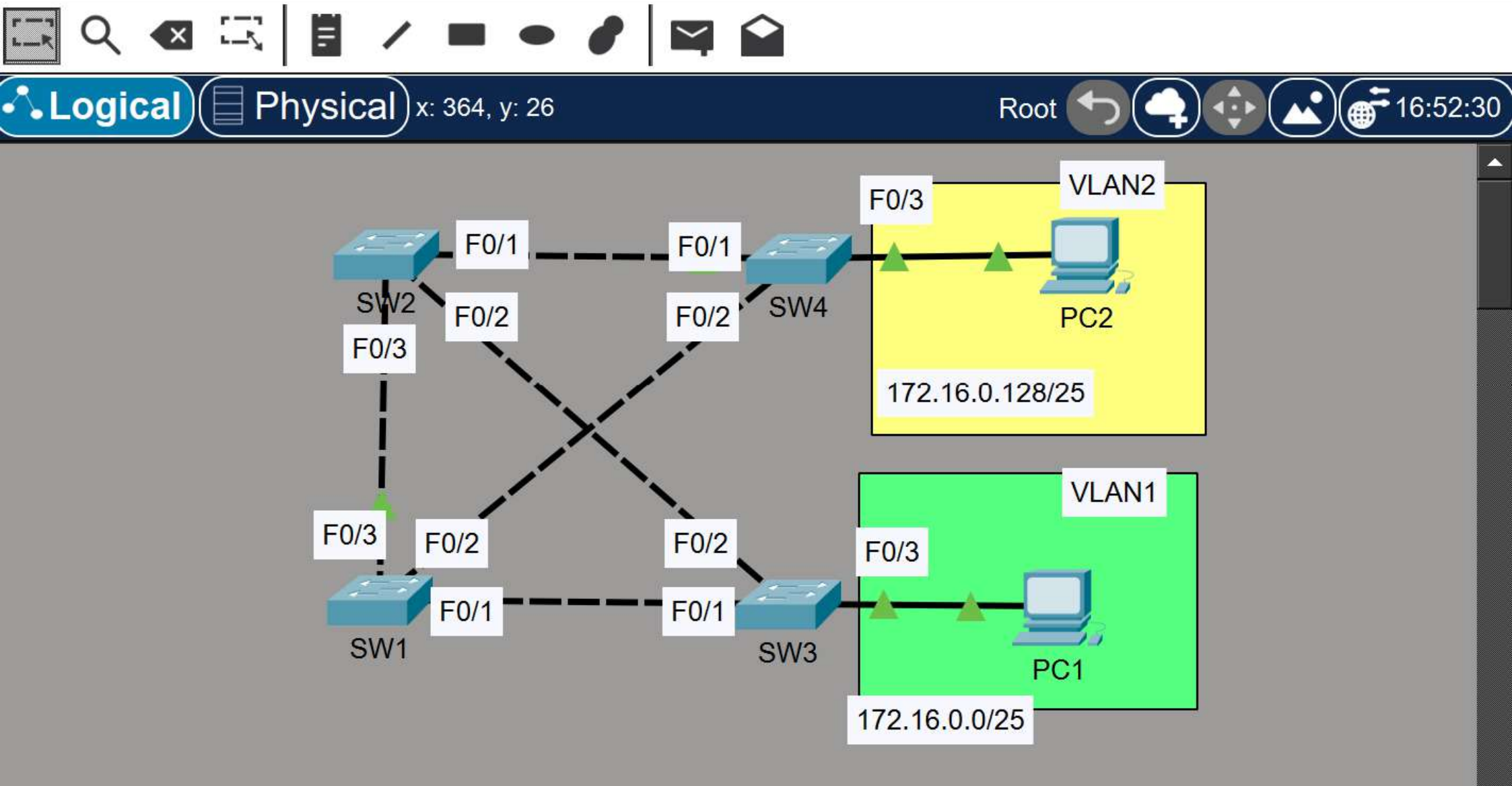


Over here I created the STP lab topology with redundant links between switches and separated end devices into VLAN 1 and VLAN 2. Here I wanted to test how PVST+ allows interfaces to maintain a different stp topology for each VLAN



Before making any changes, I verified the initial spanning-tree state using "show spanning-tree" and reviewed the root bridge, port roles, and interface statuses for both VLAN 1 and VLAN 2.

```
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    32769
           Address    0001.4301.4B81
           Cost        19
           Port        3(FastEthernet0/3)
           Hello Time  2 sec   Max Age 20 sec   Forward Delay 15 sec

Bridge ID   Priority    32769  (priority 32768 sys-id-ext 1)
           Address    0060.2F90.D14A
           Hello Time  2 sec   Max Age 20 sec   Forward Delay 15 sec
           Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Altn BLK 19        128.1   P2p
Fa0/2          Desg FWD 19        128.2   P2p
Fa0/3          Root FWD 19        128.3   P2p

VLAN0002
Spanning tree enabled protocol ieee
Root ID    Priority    32770
           Address    0001.4301.4B81
           Cost        19
           Port        3(FastEthernet0/3)
           Hello Time  2 sec   Max Age 20 sec   Forward Delay 15 sec

Bridge ID   Priority    32770  (priority 32768 sys-id-ext 2)
--More--
```

I configured SW1 as the primary root bridge for VLAN 1 and the secondary root bridge for VLAN 2. Here im putting into practice how PVST allows me to control the STP root election process for both VLANs.

```
SW1#conf t
SW1#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#sp
SW1(config)#spanning-tree vlan 1 root primary
SW1(config)#spanning-tree vlan 1 root primary
SW1(config)#sp
SW1(config)#spanning-tree vlan 2 root sec
SW1(config)#spanning-tree vlan 2 root secondary
SW1(config)#
```

I verified the STP changes on SW2 and the output showed the expected root bridge information for each VLAN.

```
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    24577
           Address    0060.2F90.D14A
           Cost       19
           Port       3(FastEthernet0/3)
           Hello Time 2 sec   Max Age 20 sec   Forward Delay 15 sec

Bridge ID   Priority    28673   (priority 28672 sys-id-ext 1)
           Address    0001.4301.4B81
           Hello Time 2 sec   Max Age 20 sec   Forward Delay 15 sec
           Aging Time 20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/1       Desg FWD 19        128.1    P2p
Fa0/3       Root FWD 19        128.3    P2p
Fa0/2       Desg FWD 19        128.2    P2p

VLAN0002
Spanning tree enabled protocol ieee
Root ID    Priority    24578
           Address    0001.4301.4B81
           This bridge is the root
           Hello Time 2 sec   Max Age 20 sec   Forward Delay 15 sec

Bridge ID   Priority    24578   (priority 24576 sys-id-ext 2)
           Address    0001.4301.4B81

--More--
```

I checked the spanning-tree output to check interface roles and path costs after the root bridge modifications. This confirmed that traffic would follow the intended forwarding paths.

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/3	Desg	FWD	19	128.3	P2p

VLAN0002

Spanning tree enabled protocol ieee

Root ID	Priority	24578
	Address	0001.4301.4B81
	Cost	19
	Port	3(FastEthernet0/3)
	Hello Time	2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID	Priority	28674 (priority 28672 sys-id-ext 2)
	Address	0060.2F90.D14A

--More--

```

SW4#conf t
SW4#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW4(config)#interface f0/2
SW4(config-if)#span
SW4(config-if)#spanning-tree vlan 1 cost 100
SW4(config-if)#exit
SW4(config)#do show spanning-tree vlan 1
VLAN0001
    Spanning tree enabled protocol ieee
    Root ID      Priority      24577
                Address        0060.2F90.D14A
                Cost           38
                Port           1(FastEthernet0/1)
                Hello Time     2 sec   Max Age 20 sec   Forward Delay 15 sec

    Bridge ID    Priority      32769   (priority 32768 sys-id-ext 1)
                Address        0090.0C03.2D70
                Hello Time     2 sec   Max Age 20 sec   Forward Delay 15 sec
                Aging Time     20

Interface                Role Sts  Cost           Prio.Nbr  Type
-----
Fa0/1                    Root FWD  19             128.1     P2p
Fa0/2                    Altn BLK  100           128.2     P2p

SW4(config)#

```

I increased the STP cost on SW4's Fa0/2 interface for VLAN 1 using "spanning-tree vlan 1 cost 100". This influenced STP path selection and caused the port to transition into a blocking state.

I reviewed the spanning-tree status on SW3 to examine the new root port and designated port assignments. This ensured the topology remained loop-free after the changes.

```
SW3>
SW3>enable
SW3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#do show span vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID      Priority      24577
               Address      0060.2F90.D14A
               Cost         19
               Port         1(FastEthernet0/1)
               Hello Time    2 sec   Max Age 20 sec   Forward Delay 15 sec

  Bridge ID    Priority      32769   (priority 32768 sys-id-ext 1)
               Address      0040.0B50.AA56
               Hello Time    2 sec   Max Age 20 sec   Forward Delay 15 sec
               Aging Time    20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Altn	BLK	19	128.2	P2p
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/3	Desg	FWD	19	128.3	P2p

```
SW3 (config)#
```


Here I tried Testing to see the effect of changing Port priority of interface Fa0/1, And nothing happened to the interface role! Port priority is only considered after the root path cost and the bridge ID have already been compared. Since those first two values were different, the port priority was never used in the election process.

```
Interface          Role Sts Cost          Prio.Nbr Type
-----
Fa0/1              Desg LIS 19          240.1      P2p
Fa0/2              Desg FWD 19          128.2      P2p
Fa0/3              Desg FWD 19          128.3      P2p

VLAN0002
  Spanning tree enabled protocol ieee
  Root ID    Priority    24578
             Address    0001.4301.4B81
             Cost      19
             Port      3 (FastEthernet0/3)
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28674  (priority 28672 sys-id-ext 2)
             Address    0060.2F90.D14A

--More--
```

aging time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----
Fa0/2	Altn	BLK	19	128.2	P2p
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/3	Desg	FWD	19	128.3	P2p

SW3(config) #